

The human L-type voltage-gated calcium channel 1.2 (hCaV1.2) plays a vital role in cardiac and neuronal excitability. Despite available cryoEM structures of hCaV1.2 and homologous channels, the C-terminal region, crucial for regulatory interactions with calmodulin and CaV channel clustering, still needs to be structurally resolved. Unlocking the complete structures of different functional states of the hCaV1.2 channel promises a better understanding of its regulation and potential therapeutic modulation. We leveraged AlphaFold 2 and AlphaFold multimer structural prediction approaches to model the hCaV1.2 channel and its protein-protein interactions. AlphaFold captured the alpha subunit of the channel in several putative conformational states. Subsequently, we investigated hCaV1.2's interactions with calmodulin as well as CaV channel clustering. Our computational structural modeling revealed the molecular basis of hCaV1.2's interactions and the dynamic multistate of calmodulin binding in various channel states, we uncovered how calmodulin binds to hCaV1.2, shedding light on channel gating and calmodulin mediated regulation. We provide a novel structural model for the dimerization of hCaV1.2 alpha subunits using AlphaFold Multimer to simulate dimerization of hCaV1.2 alpha subunits, we offer new structural perspective on the functional consequences of channel-channel interactions, potentially impacting cellular excitability and calcium signaling pathways. Our findings provide a comprehensive view of hCaV1.2's structural dynamics and contribute to a more complete understanding of hCaV1.2 channel regulation.

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Rational enzyme engineering via generative AI

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Rational enzyme engineering necessitates an in-depth understanding of various enzyme mutations, a pursuit that poses significant difficulties. However, this obstacle could potentially be overcome by learning from nature's own mechanisms of evolving efficient enzymes. We employed generative AI to analyze natural protein sequences, thereby establishing a connection between enzyme evolution and catalysis. We found that the sequence information could be used to predict enzyme activity and stability in the enzyme active center and scaffold, respectively. The discovery paves the way for a fresh understanding of both enzyme catalysis and evolution. Moreover, the enzyme evolution-catalysis relationships provide a rational way for enzyme engineering, with a ~50% success rate in enhancing enzyme performance in the experiment. Collectively, our research highlights the immense potential of the abundant natural protein sequences combined with generative AI for advancing enzymology and enzyme engineering.

Posters: Protein Stability, Folding, and Chaperones I

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Small heat shock protein HSPB1: Molecular insights into its structural complexity and chaperone activity

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Small heat shock proteins (sHSPs) serve as ATP-independent molecular chaperones, crucial for preventing aberrant protein aggregation and maintaining protein solubility. Among these, HSPB1 is a pivotal human sHSP recognized for forming large, dynamic oligomers. These oligomers' quaternary structures and chaperone activities are modulated by environmental stressors such as heat shock and acidosis. Under such stress conditions, HSPB1 oligomers have the potential to dissociate into dimers and monomers, thereby significantly augmenting chaperone activity. This transformation is underpinned by the continuous redistribution of binding partners, emphasizing the conformational dynamic of HSPB1. Despite its significance, delving into the structural aspects of HSPB1 has posed challenges, primarily due to the intrinsic disorder of its N- and C-terminal domains. In this study, we leverage AlphaFold2 modeling coupled with molecular dynamics simulations to unravel the structural foundations of HSPB1 across its monomeric, dimeric, and dodecameric forms. Further, we elucidate the multifaceted interactions between HSPB1 and its client proteins in both phase-separated and fibrillar states. Our insights underscore the indispensable role of the N-terminal domain in stabilizing diverse architectures and modulating the chaperone activity of HSPB1.

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Myo-inositol suppresses γ D-crystallin aggregation by interacting with a non-native state

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Since the aggregation of misfolded human γ D-crystallin protein in the ageing lens of the eye is implicated in cataract disease, there is interest in developing therapeutics that disrupt this process. *Myo*-inositol, a sugar alcohol that is naturally abundant in the healthy lens, was recently discovered to suppress aggregation of γ D-crystallin *in vitro* by inhibiting the formation of aggregation precursors. In our previous work, unfolding intermediates for the N-terminal domain of γ D-crystallin were proposed using Monte Carlo simulations, and *myo*-inositol was shown to preferentially bind to a persistent, early unfolding intermediate using molecular dynamics simulations. A possible binding site for *myo*-inositol on this intermediate was identified near the N-terminal β -hairpin, and previous work from our group showed that detachment of this hairpin is implicated in aggregation. By forming hydrogen bonds with both this hairpin and the rest of the domain, *myo*-inositol may keep the hairpin tethered to the domain, thereby preventing it from unfolding and the protein from aggregating. This work further characterizes the interactions between *myo*-inositol and γ D-crystallin unfolding intermediates, including the effect of binding on protein dynamics. Mutations to this binding site which may disrupt *myo*-inositol binding are identified and characterized *in silico*, for future *in vitro* experiments determining their effects on the extent of suppression of aggregation by *myo*-inositol. In addition, a series of small molecules are docked against this non-native binding site, and higher predicted binding affinities are associated with experimental measures for greater suppression of aggregation. Finally, this proposed *myo*-inositol binding site shares structural similarities with a calcium binding site found in ancestral crystallin proteins, suggesting a possible stabilizing role. Insights from this work characterizing how *myo*-inositol shifts γ D-crystallin away from aggregation-permissive states may inform the development of therapeutic candidates to treat or prevent cataract disease.

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Influence of detergents and amino acid ionic liquids on the stability of azurin

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Azurin has been utilized as a model system in biophysical investigations due to its uniquely blue-shifted fluorescence emission properties. The protein, originally derived from *Pseudomonas aeruginosa*, is a small, highly soluble protein with a mixed helical and sheet globular structure and a copper ion chelated in the core of the protein. In addition to the uniquely blue-shifted fluorescence emission spectrum, the copper in the protein structure yields a signature absorbance band in the visible region allowing for an additional probe of protein folding. Ionic liquids (ILs), often referred to as room-temperature ionic liquids or molten salts, have garnered great interest in the last 15-20 years as potential components of electrochemical devices or applications. More recently, the biocompatibility of these molecules has developed an increased interest in the field, especially considering that some IL species can stabilize biomolecular structures while other ILs strongly destabilize 3D structures. In this study, we used absorbance and fluorescence spectroscopy to characterize the unfolding of myoglobin and the impact amino acid-based ILs had on this process. The IL cations explored in this study are tetramethyl guanidine and choline, both selected due to the high biocompatibility of these molecules. The amino acid anions of the IL pairs were selected as serine. Serine identified from previous studies showing strong destabilizing effects on the protein azurin when denatured using thermal unfolding approaches. In this study we coupled the aa-ILs with detergent-mediated denaturation using a cationic (TTAB), zwitterionic (Empigen BB), and anionic (SDS) detergent. The denaturation process was monitored by absorbance spectroscopy arising from the copper and Trp fluorescence of the azurin. Notably, the anionic detergent SDS had significantly less impact on the denaturation of the protein than the cationic and zwitterionic detergents.